Multiscale Location Attention Network for Building and Water Segmentation of Remote Sensing Image

Xin Dai, Min Xia, *Member, IEEE*, Liguo Weng, Kai Hu, Haifeng Lin, and Ming Qian

***Abstract*— Traditional building and water segmentation methods are vulnerable to noise interference, and hence, they could not avoid missed and false detections in the detection process. Excessive deep learning downsampling would lead to significant loss of feature map information, image location information offset, and the overall effect of falling apart. To address these issues, a multiscale location attention network (MSLANet) is proposed. Location-spatial information and channel information are particularly important for edge detail segmentation in building and water cover. The network includes a location channel attention (LCA) unit to focus on tributary details of rivers and segmentation of building edge eaves. Moreover, this article builds a dual-branch multiscale aggregation (DBMSA) unit to obtain deeper multiscale semantic information. Finally, the multiscale fusion unit (MSF) is used to guide the information merging of multiple stages, and the boundary information is improved by splicing the acquired deep multiscale information with the information of the relevant feature extraction layer in the downsampling. The experimental results on several datasets show that the proposed approach outperforms other methodologies in segmentation accuracy.**

***Index Terms*— Attention, building and water, deep learning, image segmentation.**

I. Introduction

BY ANALYZING building and water datasets, we could understand urbanization rates, water cover rates, and other environmental changes. The rapid development of high-resolution remote sensing satellite technology challenges the accuracy of land surface classification [1]. In recent years, the development of new technologies, such as artificial intelligence, has provided a way to achieve automatic image segmentation [2]. In remote sensing images, water bodies and buildings are very important landmarks, and they have important practical significance in land planning, land classification, urbanization speed research, water resources management, and water resources protection [3]. Therefore, accurate segmentation of water and buildings is required.

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Currently, with the wide application of convolutional neural network (CNN) in image processing, high-resolution remote sensing image segmentation technology has ushered in an innovation [4], [5], [6], [7]. At first, deep convolutional networks were used for image classification. Compared with ordinary classification tasks, building and water segmentation is a pixel-level classification assignment, so ordinary image classification networks cannot accurately complete the segmentation task. To perform pixel-level segmentation, Long et al. [8] presented fully convolution network (FCN) with visual geometry group (VGG) as the backbone network, and it extracted the final features through a deconvolution layer. The layer is upsampled to output an accurate image segmentation. Ronneberger et al. [9] presented a network called UNet that inherits the idea of the FCN symmetric codec structure. It uses the skip connection method to create the feature extraction layer and the upsampling layer. Feature fusion improves segmentation accuracy. The DeepLab series of networks proposed by Chen et al. [10] use atrous convolution and conditional random field methods to improve prediction results. Among them, the atrous spatial pyramid pooling (ASPP) module in the DeepLabV3Plus network achieves a larger receptive field without losing too much image resolution, which improves the image segmentation accuracy. Zhao et al. [11] proposed pyramid scene parsing network (PSPNet), and the network uses the pyramid pooling module (PPM), and it could combine relevant data from several places, boosting its capacity to gather global information.

The above networks are all image semantic segmentation networks. For specific remote sensing image segmentation, the accuracy is not high enough. To address these issues, Chen et al. [12] proposed a multilevel attention upsampling module, and it can use high-level features to provide guidance information for low-level features to generate new features, optimize the upsampling process, and improve segmentation results. Its disadvantage is that this module only uses global pooling and convolution operations, and it pays too much attention to mining the local information of the image, ignoring the global, and hence, it is easy to cause missed detection and false detection. Huang et al. [13] proposed dual function feature aggregation network (DFFANet), which aggregates the contextual information of the image by using a self-attention-like mechanism. This method can effectively eliminate the background information and pay more attention to the target area. However, its disadvantage is that the network can only capture the overall characteristics of the segmentation target,

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and it ignores the importance of edge details. Liu et al. [14] constructed a dense dilated convolutions merging network for building and water classification, but its segmentation effect is still not good. Gao et al. [15] proposed multichannel feature fusion Lozenge network (MLNet), which compresses global information through a weighted exponential attention mechanism and restores the original information according to different weight allocation ratios during the upsampling process. The advantage of this method is that the weights are reasonably allocated to obtain the optimal segmentation image through the weight index attention mechanism. Hu et al. [16] proposed the deep feature extraction (DFE) module. The advantage of this method is that it uses location attention. The location information is obtained through the pooling layer to encode the feature map, which enhances the network's attention to the target. The disadvantage is that the position attention generated by the pooling layer is not capable of capturing the long-term distance dependence of the image, which is easy to cause false detection and missed detection of the target object.

Although the abovementioned networks perform well on many semantic segmentation tasks, they have deficiencies in building and water segmentation [17], [18]. First, the classical semantic segmentation network has only a simple feature extraction process and does not incorporate an effective attention module, which leads to the lack of the ability to capture detailed information. Because the network for building and water segmentation uses too many downsampling processes, the info of the feature map is seriously lost, the position information of the image is shifted, and the whole image is divided, which will cause misjudgment when facing objects with similar pixel values. For example, it is easy to misidentify a house with a blue roof as a water area. This is because the network does not accurately extract semantic information, and the abovementioned situations will cause the wrong segmentation. To address the aforementioned issues, this research proposes a multiscale location attention network (MSLANet). Fig. 1 depicts the network's general structure. The network uses the improved ResNet50 [19] as the backbone to solve the problem of gradient disappearance and gradient explosion caused by deep downsampling, so that the network can better extract feature information during downsampling. This article proposes a dual-branch multiscale aggregation (DBMSA) unit to achieve multiscale feature aggregation to improve the receptive field, acquire multiscale background information, and improve the segmentation capacity of buildings as well as water of multiple scales without increasing the calculation parameters. In addition, the local channel attention unit constructed in this article is used for skip connections. It enhances the network's representation of objects of interest by extracting attention features both vertically and horizontally in the feature map. This enables the network to encode spatial information and enables attention module to more accurately locate the exact location of objects of interest. The multiscale fusion (MSF) unit plays the role of guiding the info interaction between deep features and shallow features, so that the two can guide each other in the upsampling process. This module integrates the rich semantic information and vast spatial information in the image, which can improve the overall segmentation. Finally, the two 3×3 thinning convolutions are refined to complete the edge details of buildings and rivers again.

Compared with existing attention-based remote sensing image segmentation methods, our model has the following advantages. First, this work innovatively proposes a method that combines global pooling convolution operations with nonlocal neural networks. This method can input the perceptual features in the vertical direction and the horizontal direction obtained through the global pooling convolution into the nonlocal neural networks. This operation can guide the location-spatial attention (LSA) unit to further obtain the long-distance dependencies of the feature maps along the H- and W-directions. Afterward, we summarize the feature information in the H- and W-directions through a weighting operation to complete the position information of the feature map. The advantage over nonlocal neural networks module is that this attention module can capture long-range dependencies along one spatial direction and preserve precise location information along another spatial direction. In addition, it can capture direction-aware and location-sensitive information. This helps the model locate and identify objects of interest more accurately. Besides that, we embed location information into channel information. It can make full use of the captured location information to accurately highlight the region of interest. At the same time, it can also effectively capture the relationship between channels, generate effective weighted features, and enhance the feature representation ability of the network. Second, our proposed DBMSA unit employs a set of convolution kernels with different sampling rates to convolve the feature maps. This method increases the receptive field of the convolution operation, thereby realizing the extraction of features of different scales. At the same time, we added depthwise separable convolutions. It can effectively reduce the amount of parameters and calculations of the model and maintain a high accuracy. By converging features at different scales, the network obtains a global feature representation. In addition, the module adds a residual connection, which can fuse the features of different scales with the original unprocessed feature information to complement the feature information of the image. The multiscale feature information obtained through the DBMSA unit can improve the model's ability to perceive buildings and water of different sizes. At the same time, this module can also make the model more robust and generalizable. Finally, the MSF unit proposed in this article adopts the idea that low-level detail information and high-level semantic information could guide each other. High-level features contain a large amount of category information, which guides the classification of low-level features. Low-level features contain important detail information, which can enhance the detailed recognition of buildings and water. This module enhances the information interaction between channels by guiding different scales. At the same time, the thinning convolution module at the output completes the final restoration of the edge of the feature map during the upsampling process. Experiments show that the method proposed in this article has great advantages compared with other existing attention-based networks. The proposed model in this work has made the following contributions.

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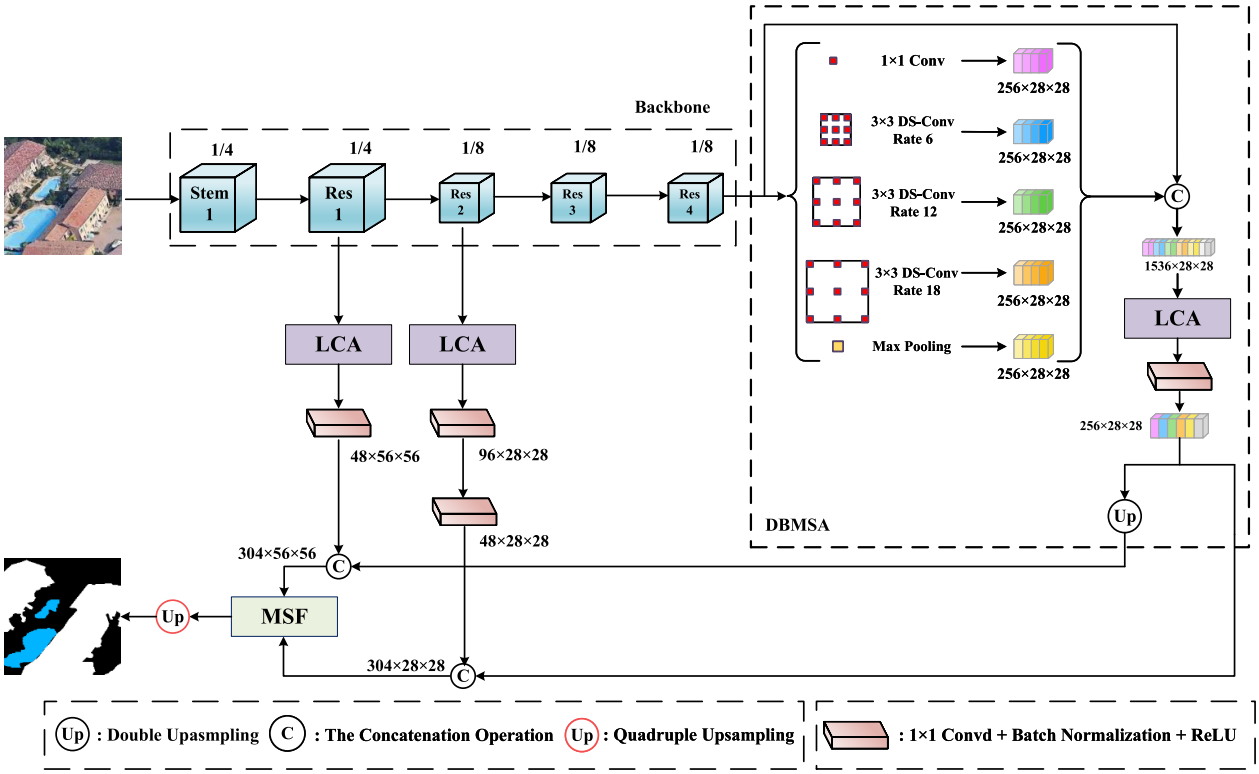

Fig. 1. Framework of the MSLANet. The model consists of a basic network and three auxiliary modules. The auxiliary modules include DBMSA module, MSF module, and LCA module.

1) A location channel attention (LCA) unit is designed to embed location information into CA and spatial attention and aggregate features along two spatial directions (length and width), respectively. It can not only focus on capturing the long-range dependencies in one direction of the feature map, but also retain the precise pixel position information in this direction. This design enhances the network's capture of objects of interest and enhances its focus on rivers and building edges. The location global attention module realizes the capture of long-distance dependencies between pixel locations and global information. The CA module also effectively pays attention to the edge information of buildings and water and obtains effective detailed information. The network can achieve accurate segmentation of water and buildings by combining local information with global information.
2) In order to achieve accurate segmentation of water and buildings, it is necessary to combine multiscale image features, so a DBMSA unit is designed, and it adds atrous convolution to expand the receptive field and improves multiscale semantic information extraction ability. The output of the dual branch can be combined with LCA unit and MSF unit to achieve multilevel features exchange.
3) An MSF unit is proposed, which can effectively fuse the semantic features of high-level feature maps with the spatial information of low-level feature maps and complete the information in the upsampling stage. At the same time, two 3×3 thinning convolutions can optimize the network's ability to capture water and building details, making the boundaries clearer.

II. Methodology

Previous research networks show poor segmentation results because of the negligence of details. Aiming at the problems of easy misclassification and rough segmentation boundaries in building and water segmentation, the MSLANet is proposed. It can quickly and effectively segment buildings and water. Fig. 1 depicts the whole network design. We initially employ ResNet50 as the backbone network to extract features at various stages, and then, we use a DBMSA unit to extract multiscale features. The encoding portion of the entire network is used to capture deep semantic features, which consist of the above two structures. At the same time, the LCA unit processes the feature maps in the encoding stage at the skip connections, which strengthens the network's capture of objects of interest. The proposed MSF unit may lead the network for information interaction and refine the semantic and spatial properties of the feature map during the decoding stage. Finally, the picture formed after fusion is refined and convoluted to reidentify the target border and increase boundary segmentation accuracy.

A. Backbone

We use the improved ResNet50 as the backbone network, and its residual structure solves the problems of gradient

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TABLE I
ARCHITECTURE OF ORIGINAL AND MODIFIED RESNET50

	Original		Modified	
Layer	50-layer	Size	Modified 50-layer	Size
Stem	7×7,stride 2	1/2		1/2
L1	3×3,Max pool,stride 2 $\begin{pmatrix} 1\times1,64 \\ 3\times3,64 \\ 3\times3,64 \end{pmatrix}$	1/4		1/4
L2	$\begin{pmatrix} 1\times1,128 \\ 3\times3,128 \\ 3\times3,512 \end{pmatrix}$	1/8		1/8
L3	$\begin{pmatrix} 1\times1,256 \\ 3\times3,256 \\ 3\times3,1024 \end{pmatrix}$	1/16	Dilated Convolution	1/8
L4	$\begin{pmatrix} 1\times1,512 \\ 3\times3,512 \\ 3\times3,2048 \end{pmatrix}$	1/32	Dilated Convolution	1/8
Average pool, 1000-d fe, Softmax				

explosion and gradient disappearance that are when the convolutional network is too deep very common [20]. The improved ResNet50, as the backbone network of the coding structure, can perform feature extraction at different levels, but since it was originally used for classification, it needs to be modified to solve the building and water segmentation task. First, the last fully connected layer of ResNet50 is removed, and the feature extraction layer of the backbone network is divided into five layers. Except for the first layer, each of the remaining layers consists of several residual blocks, which can be described as follows:

$$x_i^{'} = \mathrm{Conv}_i^2\left[\sigma\left(\mathrm{Conv}_i^1(x_i)\right]\right. \tag{1}$$

$$x_{i+1} = x_i + \sigma\left(\mathrm{Conv}_{i+1}^1\left\{\sigma\left(x_i^{'}\right)\right\}\right) \tag{2}$$

where x_i is the matrix input, $x_i^{'}$ is the output of the intermediate block, x_{i+1} is the output of the residual block Conv_i^1, Conv_{i+1}^1 represents the convolution operations with a convolution kernel of 1, Conv_i^2 represents the convolution operation with a convolution kernel of 3, and $\sigma(\cdot)$ represents the BatchNorm and the activation function rectified linear unit (ReLU). Based on ResNet50, this article designs a new structure for building and water feature extraction. For this network, there are two main improvements. First, we add atrous convolution to the fourth and fifth layers of the feature extraction layer. Atrous convolution can increase the map area without losing features. Second, we remove the terminal structure of the original network and modified the feature map size of the fourth and fifth layers in order to match the input size of the third layer, so that ResNet50 is improved to have eightfold downsampling only, as shown in Table I.

B. DBMSA Unit

In the buildings and water datasets, the eaves and chimneys of the houses in the images selected in this article are small, and most of them overlap with the main body of buildings. Therefore, we classify house edges and chimneys as classes of buildings. In the segmentation task of buildings and water, the identification of small tributaries of water and detailed information of building edges is very important. Because the abovementioned feature information is difficult to obtain, in addition to capturing detailed information in low-level downsampling, we also need to mine deeper semantic information [21]. During the upsampling process, the deep semantic information has a guiding effect on patching the low-level detail information. High-level semantic information can provide contextual information of the entire image or region, helping the model to better understand local details. At the same time, in semantic segmentation, high-level semantic information usually includes information, such as object categories and boundaries. This information can help the model better understand the shape and structure of objects, thus better segmenting objects and enhancing segmentation boundaries. Although this article chose ResNet50 as the backbone network, changed the feature extraction layer of the last two layers, and added atrous convolution, its ability to extract deep semantic information is still limited. In order to extract high-level semantic information and improve the ability to obtain global information, Zhao et al. [11] tried to use the PPM, which can aggregate the context information of different regions and is very effective for processing large-scale venues and lakes. However, it is not very effective for small water. A large convolution kernel will extract a lot of irrelevant information and interfere with the prediction of the model. In addition, the number of output channels of this module is too large, resulting in excessive calculation parameters and low efficiency.

The DBMSA unit proposed in this article uses a set of convolution kernels with different sampling rates to convolve the feature map. As shown in the black dotted box in Fig. 1, the DBMSA unit includes three depthwise separable convolutions with different dilation rates with a kernel of 3, a convolution with a kernel of 1, and a global max-pooling operation. This block can be used to obtain multiscale information and deep semantic information. When using convolution kernels with different sampling rates, DBMSA unit introduces holes of different sizes on the feature map to change the receptive field of convolution, thereby realizing the extraction of different scale features of water and buildings. At the same time, the model in this article also adopts the idea of pyramid pooling to aggregate features at different scales. After the features of different sizes are pooled, the network obtains a global feature representation, which further improves the accuracy of the model. Too many channels of input features will lead to a large amount of parameter calculation and low efficiency. Therefore, we incorporate depthwise separable convolutions. It can effectively reduce the number of parameters and computation of the model while maintaining a high accuracy. In addition, the module adds a residual connection. This operation can fuse the features of different scales with the original unprocessed feature information to complement the feature information of the image. After multiscale extraction, the feature information is converted and convoluted through the channel with a convolution kernel of 1, which reduces

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the number of channels and reduces the computational load of the entire module. After channel reduction, the extracted multiscale information is further optimized by a local CA unit. Afterward, the extracted multiscale feature maps and feature maps with added attention are input into the channel conversion convolution to further reduce the number of channels. Finally, the processed feature information is connected to the feature information of the jump connection through dual output branches, so as to improve the exchange and fusion of low-level spatial information and high-level semantic information, and complete multiscale extraction.

C. Location CA Unit

Attention mechanisms [22], [23] are now widely used in various computer vision tasks [24]. The existing attention modules, such as convolutional block attention module (CBAM) and bottleneck attention module (BAM), first reduce the channel dimension of the input tensor, and then use convolution calculation to extract the channel information and spatial information of the feature map. This method can effectively enhance the network's attention to important objects in the image. However, convolution can only capture local relations, its ability to capture global spatial information is weak, and there is no way to complete the long-term distance dependencies necessary in semantic segmentation. In contrast, the DFE module proposed by Hu et al. [23] adopts positional attention, which obtains positional attention through avg-pooling operation to obtain global information. However, the ability of the pooling layer to obtain the long-term distance dependency of the image is not strong, and the simple pooling operation is still prone to false detection and missed detection of the target object. Therefore, this article proposes the location CA unit to improve the nonlocal neural networks module, so as to make up for the defect that the convolution operation cannot obtain global information. The LCA unit is regarded as a element that enhances the expressiveness of the network learned features.

The location CA module in this article is divided into LSA and CA. Due to the diversities of buildings and water, there are some hollowed-out buildings in the middle of the buildings and observation towers in the pools, which will cause misclassification of the network. Besides, the roofs of some buildings, such as blue-top containers, have similar properties to water in the visible spectrum, which can cause misclassification because of the lack of location information and global information. In the encoding stage, the network obtains enough semantic information, but the underlying feature map loses a lot of global information due to multiple downsampling and convolution operations, which causes the network to lose large-scale features of water or buildings. As a result, buildings and water with blue roofs are misclassified. This article develops an LCA unit (as illustrated in Fig. 2) to achieve a more precise categorization. This module innovatively uses the combination of nonlocal neural networks and global pooling to extract the spatial position information of the image and can also embed the position information into the channel information. This method can strengthen the global attention

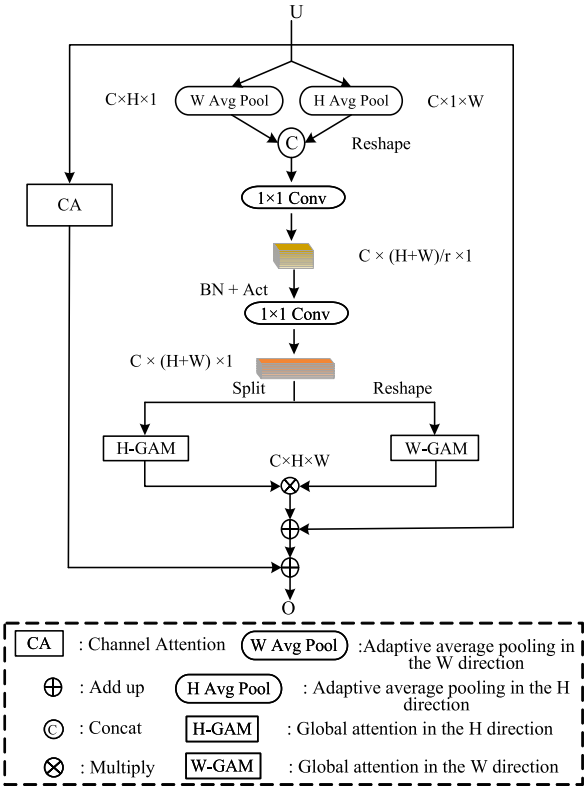

Fig. 2. Structure of location CA unit.

module, enable the network to further mine information along the horizontal and vertical directions of the feature map, and capture the long-distance dependence of the input feature map along one direction. In the location information, LCA enhances the network's attention to the hollow edge of the house, the edge of the swimming pool at the top of the house. In the global information, LCA centrally obtains the overall characteristics of large factories and container houses with blue roofs, so as to achieve excellent segmentation.

Meanwhile, inspired by the attention modules squeeze and excitation (SE) module [25], CBAM module [26], selective kernel (SK) module [27], and coordinate attention [28], we construct a new CA unit in LCA, as shown in Fig. 3; it uses two pooling layers, global average pooling and global max pooling, to extract high-level features. After the feature information passes through two global pooling layers, the channel conversion is performed through the convolution layer with the kernel size of 1×1. Due to the excessive number of network channels in the downsampling process, we set a channel reduction value R in the position CA module to reduce the number of channels. After experimental comparison, we found that when the channel reduction value R is set to 8, the performance of the network is the best. The operation formula is as follows:

$$x_{\max} = \mathrm{Conv}(g_{\max}(x_i)) \tag{3}$$

$$x_{\mathrm{avg}} = \mathrm{Conv}\left(g_{\mathrm{avg}}(x_i)\right). \tag{4}$$

Among them, $g_{\max}$ and g_{avg}, respectively, represent global maximum pooling and global average pooling, and Conv represents the convolution operation with a kernel 1.

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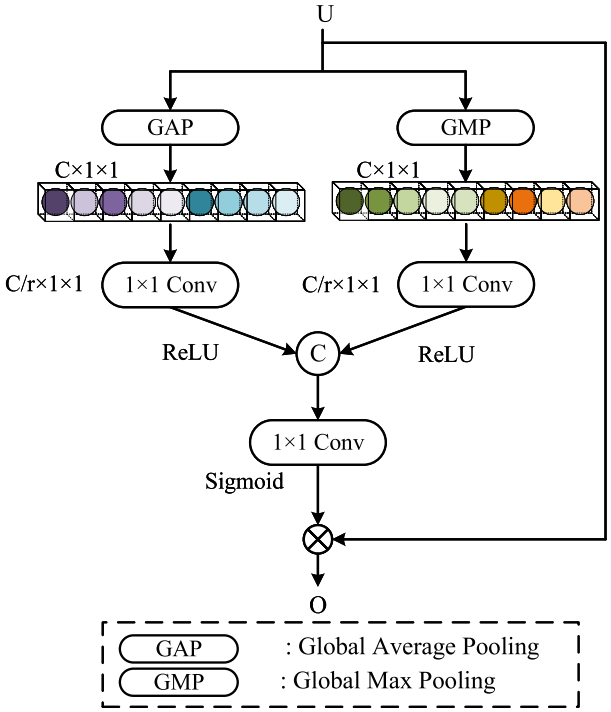

Fig. 3. Structure of CA unit.

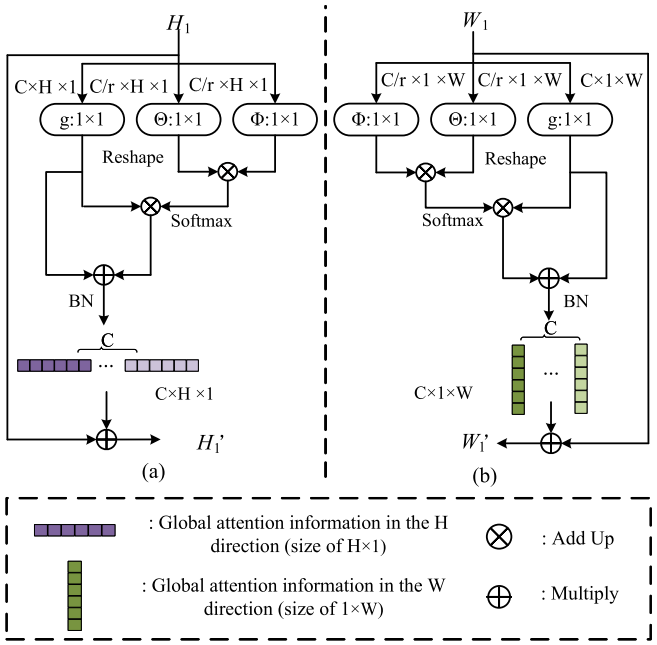

Fig. 4. Structure of LSA unit.

Then, through the above operations, we spliced the outputs of the two branches to obtain new feature information. Next, a convolution with kernel 1 is used as the selector to adaptively focus on the two globally pooled feature representations. By this method, the attention to the pixel color of buildings and water edges is increased, and the interference of useless information is mitigated. Finally, the original feature map is reweighted using the nonlinear activation function Sigmoid, which is formulated as follows:

$$y_{\text{out}} = x_i \cdot \text{Sigmoid}\left(\text{Conv}\left(\text{Cat}\left(\lambda_{\max}, \lambda_{\text{avg}}\right)\right)\right) \tag{5}$$

where Cat represents the splicing of two global pooling outputs, Conv represents the selector, Sigmoid represents the activation function, and x_i represents the input feature.

An LSA unit, which aggregates features along two spatial directions (i.e., length and width), respectively, is shown in Fig. 4. Positional attention decomposes the input features into two 1-D feature encoders through an adaptive pooling layer to effectively integrate spatial information into the generated attention feature maps. Traditional global pool compresses global spatial features into channel descriptions, making it impossible to save pixel position information, which is necessary in capturing spatial structure.

In this way, we can focus on capturing long-range dependencies along one direction, while preserving the precise pixel location information in that spatial direction. The resulting feature maps could be encoded into edge location-sensitive attention maps, respectively. The attention map is used in conjunction with the input picture to improve the network's attention to the edges of hollowed-out buildings, residences, and pools. This structure could improve segmentation accuracy and avoid overly rough edges of recognized objects by the network. The following equations can be used to describe this method:

$$x_{i+1} = \text{Conv}\left\{\sigma\left(\text{Conv}\left[\text{Cat}\left(x_{\text{avg}}(x_i), y_{\text{avg}}(x_i)^T\right]\right)\right\} \tag{6}$$

where x_i indicates the input feature, x_{avg} indicates the adaptive average pooling of x_i along the H-direction, y_{avg} represents the adaptive average pooling of x_i along the W-direction. y_{avg}^T means to transpose y_{avg} to facilitate subsequent fusion operations, Cat means to fuse x_{avg} and y_{avg} to gather feature information, $\sigma(\cdot)$ means to perform BatchNorm, and the activation function is ReLU.

The feature maps after the above operations will be split and then sent to the global attention modules in the H- and W-directions, respectively. The global attention module can enhance the network's ability to use long-term distances and improve the network's control over global information. Ordinary convolution operations can only obtain local information and cannot avoid miscalculations of water and houses with blue roofs. Therefore, this article improves the global attention module [29] and integrates the position information into the global attention module. This module allows the feature information to capture the global information of buildings and water along the H- and W-directions (height and width), obtain large-scale features of water and buildings, and avoid misclassification of water. Since the number of channels is too large, we reduce the number of channels of the two position elements Θ and Φ in the global attention module. After experiments, it is found that when reduction rate is 8, the network performs the best. Take the positional attention in the H-direction as an example. First, Θ and Φ perform channel merging and transposing operations, respectively, and then, matrix multiplication is performed. Then, their calculated the results would go through Softmax. Finally, their outputs are matrix multiplied with g after channel changes. Through the operation between the three position elements,

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we can realize the interaction of position features and reweight the feature map after global attention through the Softmax operation. Next, we add this weighted feature information to g together with the original feature map information. This operation completes the information, thereby could capture the long-term distance dependence of the feature map, and the precise information in the H-direction is preserved. The position attention operation in the W-direction is also the same. Finally, we aggregate the position information in the H- and W-directions and simultaneously obtain the long-term distance dependencies of the two directions. In addition, the precise pixel position information in the two directions is also retained. Afterward, the feature map that aggregates the above information is combined with the original input feature map to form a complete residual LSA module. The above method can be expressed by the following formula:

$$H_{\text{out}} = \sigma\left(\text{Conv}_g^1(h_i) + \left\{\text{Soft max}\left[\left(\text{Conv}_\Phi^2(h_i) \otimes \text{Conv}_\Theta^{2^T}(h_i)\right]\right.\right. + \left.\left.\text{Conv}_g^1(h_i)\right\}\right. \tag{7}$$

$$W_{\text{out}} = \sigma\left(\text{Conv}_g^1(w_i) + \left\{\text{Soft max}\left[\left(\text{Conv}_\Phi^2(w_i) \otimes \text{Conv}_\Theta^{2^T}(w_i)\right]\right.\right. + \left.\left.\text{Conv}_g^1(w_i)\right\}\right. \tag{8}$$

$$X_{\text{out}} = H_{\text{out}} \otimes W_{\text{out}} \tag{9}$$

$$X_{\text{location}} = X_{\text{out}} + X_i \tag{10}$$

where h_i indicates the input attention feature map in the H-direction, w_i indicates the input attention feature map in the W-direction, Conv^1 and Conv^2 indicate the convolution operation with a convolution kernel of 1, the difference lies in the number of channels, Softmax indicates the activation function Softmax, $\sigma(\cdot)$ indicates the normalization function BatchNorm, $\otimes$ indicates the dot product, X_{out} indicates the summary information in the H- and W-directions, X_{location} indicates the output of the LSA unit, and X_i indicates the input feature map of the module.

Finally, the information of the LSA unit and the CA unit is merged to obtain the final output features. This feature not only has the ability to detect detailed information, such as building chimneys and drain edges, but also could obtain global information about factories and large lakes from the long-term distance relationship. In addition, the square between the buildings and the floor tiles of the villa are similar to the building itself in terms of pixels. This situation is easy to cause misclassification of the network, and the network will classify these backgrounds as buildings, which reduces the accuracy of segmentation. The above problems can be well solved by our position CA module, and better segmentation could be obtained.

D. MSF Unit

The recognition of eaves margins and intricate tiny rivers is critical in building and water segmentation. Most current feature fusion methods use broadcast mechanism for simple dimension splicing or addition. This method will lead to the loss of the diversity of low-level spatial information and high-level semantic information.

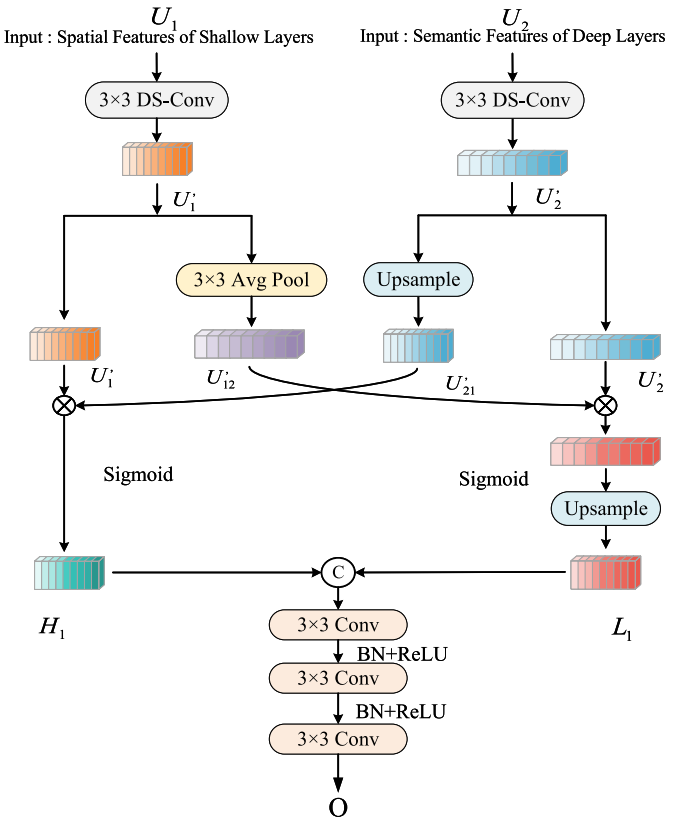

Fig. 5. Structure of the MSF module.

Therefore, this article designs an MSF unit, as shown in Fig. 5. Through the guidance of different scales, the information interaction between channels is enhanced. High-level features contain much category information, which guide the classification of low-level features, and low-level features contain important spatial features that can enhance the global recognition of buildings and water. The parameter calculation of deep networks is relatively large, and we need to perform parameter reduction. Inspired by Chollet [30], we use depth-wise separable convolution (DSConv) to mitigate the problem.

Specifically, first, 3×3 DSConv is used to obtain $U_1^{'}$ and $U_2^{'}$, which strengthens the semantic information features. For high-level features, bilinear interpolation method is used to sample it to the same size as the low-level feature map $U_{21}^{'}$. The low-level features are then weighted to get H_1 to help the lower-level feature maps enhance semantic information. In the meantime, the low-level information is downsampled to $U_{12}^{'}$ and weighted to the high-level feature map to enhance the spatial information.

Then, the guided high-level features and low-level features are fused in the channel dimension, and better performance could be achieved by increasing the number of channels, which is beneficial to increase the features describing the image itself. The expression is as follows:

$$U_1^{'} = \text{DSConv}(U_1) \tag{11}$$

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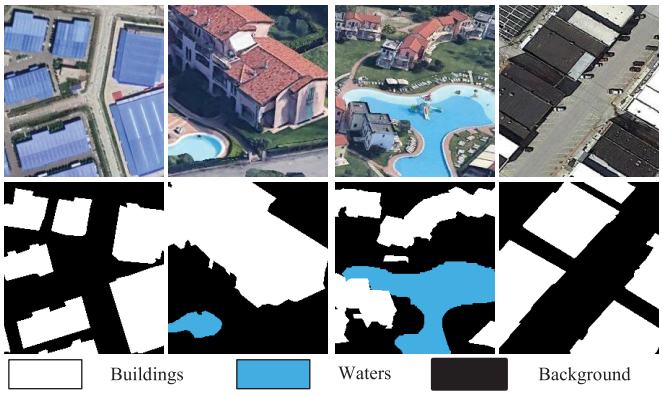

Fig. 6. Building and water datasets.

$$U_2^{'} = \mathrm{DSConv}(U_2) \tag{12}$$

$$U_{12}^{'} = \mathrm{Avg}_{3\times3}\left(U_1^{'}\right) \tag{13}$$

$$U_{21}^{'} = \mathrm{Upsample}\left(U_2^{'}\right) \tag{14}$$

$$H_1 = U_{21}^{'} \otimes \mathrm{Sigmoid}\left(U_1^{'}\right) \tag{15}$$

$$L_1 = \mathrm{Usample}\left[U_{12}^{'} \otimes \mathrm{Sigmoid}\left(U_2^{'}\right)\right] \tag{16}$$

where DSConv is a 33 DSConv operation, $\mathrm{Avg}_{3\times3}$ is a 3×3 avg-pooling operation, Upsample is a double upsampling operation, $\otimes$ is a dot product, U_1 is a low-level feature, U_2 is a high-level feature, and L_1 is the high-level feature of $U_2^{'}$ that is twice upsampled.

III. Experiments

A. Datasets

1) Building and Water Dataset: This work creates a dataset to train and test the model's effectiveness. The dataset comes from **Google Earth** maps with a size of 1600×900, which includes rural parks in Europe, coastal residential areas in China, private villas in North America, and so on. This dataset has a large dimension span, various image selection angles, and a complex background, so it has high requirements for segmentation algorithms. This article selects 300 pictures with the original size of 1600×900 from Google Earth and cuts them according to the size of 224×224. We checked the dataset obtained after cutting and removed images and labels that only had a single classification. We obtained a total of about 3200 images. After that, we performed data augmentation on these images to obtain a total of 10 000 pairs of images and labels containing buildings and water. There are three sorts of data augmentation strategies: center rotation (50%), horizontal and vertical rotation (50%), random rotation ($-10°$ to $10°$), and so on. We selected 4/5 of the images as the training set and validation set and 1/5 of the images as the test set. These images are labeled as three classes of objects: water, buildings, and background. Fig. 6 illustrates the photographs and their labels.

This dataset has the following characteristics.

1) As shown in Fig. 6(a) and (d), the top of the container house and some sundries in the image have similarity in pixel values with the water and the background. It causes misclassification easily and could test the model's segmentation ability.
2) As shown in Fig. 6(a) and (b), the houses of the two colors are taken from different angles. This way of shooting reflects the diversity of the dataset. In addition, the shooting method in Fig. 6(b) has a certain angle, so the shadows produced by the buildings will also affect the judgment of the network. This situation further tests the segmentation ability of the network.
3) The scenarios in the building and water segmentation are very diversified, including villas, factories, villages, lakes, oceans, and other scenes; it increases the dataset's authenticity and practicability and may professionally depict the model's detection ability.
4) We use manual annotation, so the information of the segmented area is more accurate.

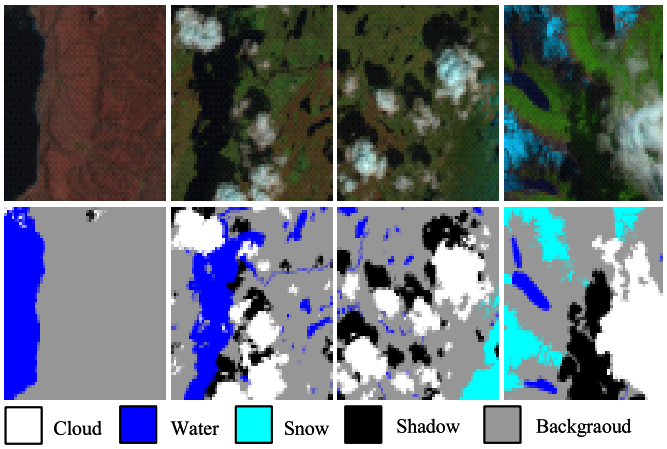

Fig. 7. L8SPARCS dataset.

2) Landsat-8 Spatial Procedures For Automated Removal of Cloud and Shadow (L8SPARCS) Dataset: This dataset contains 80 1000×1000 pixel subsets of **Landsat 8** operational land imager/thermal infrared sensor **(OLI/TIRS)** scenes [31]. Each scene comes with manual cloud-realistic masks and color composite preview images. Each image is provided as a thematic raster, including categories such as clouds, cloud shadows, snow/ice, water, and background. We divided 80 scene pictures into smaller pictures with a size of 256×256 and performed data enhancement operations on them at the same time. After this, we selected 4/5 images as training and validation sets and 1/5 images as test set. The dataset has five classification categories: background, water, snow, cloud, and cloud shadow, which can effectively test the network's segmentation performance. Some labels and pictures of this dataset are shown in Fig. 7.

3) Water Dataset: In this article, the remote sensing images of China's **HJ-1A (HJ-1B)** multispectral environmental remote sensing satellite and National Aeronautics and Space Administration (NASA) multispectral remote sensing satellite **Landsat-8** are selected as generalized datasets. After the 250 water images are divided into 256×256 images each, we obtained a total of 2800 original water images. Afterward,

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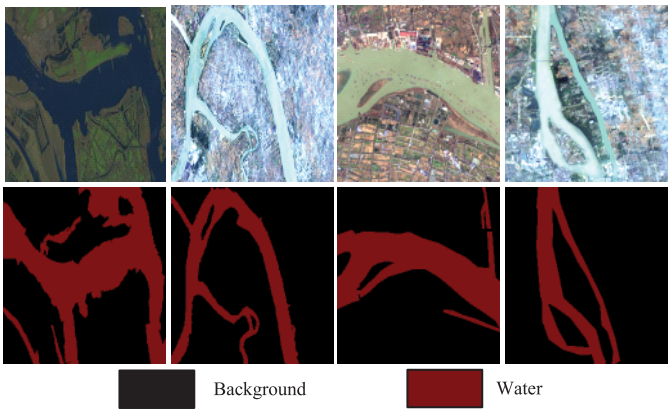

Fig. 8. Water dataset.

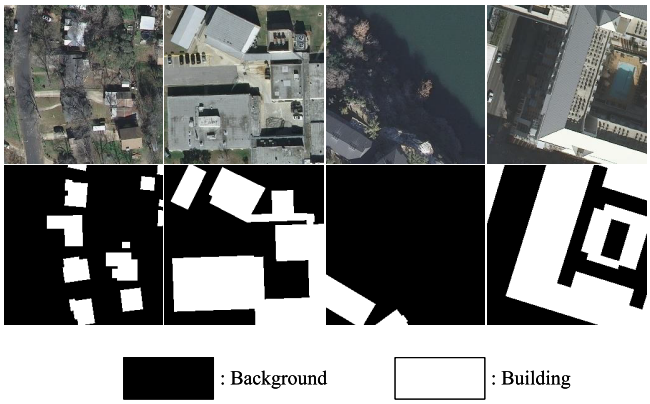

Fig. 9. Inria dataset.

we performed data augmentation on the experimental images, generating a total of 8000 pairs of water images. We selected 4/5 of the images as the training set and validation set and 1/5 of the images as the test set. The dataset has two classification categories: water and background, which can effectively test the network's performance in detecting water in various backgrounds. Fig. 8 depicts the pictures and labels from a subset of this dataset.

4) Inria Dataset: The Inria Aerial Image Labeling dataset is a subset of the Inria Dataset buildings dataset for building segmentation. The dataset has 360 aerial images with a size of 5000 × 5000. There is a training set of 180 images with corresponding labels. These aerial images come from different cities and regions, including different types of areas, such as urban, suburban, and rural. The test and training sets contain the same number of images, but the distribution of images is completely different from the training set, which also has 36 images per region. It is worth noting that this dataset does not have a separate validation set, and the corresponding label files for the test set images are not publicly available. Therefore, we cut the 180 large-scale images in the training set into images with a size of 256 × 256, and we eliminated images and labels with only a single classification. We selected 4/5 images as training and validation sets and 1/5 images as test set. These images are labeled as two types: buildings and backgrounds. Fig. 9 depicts pictures and labels from a subset of this dataset.

B. Experimental Parameter Setting

All of the tests are configured with Intel Core i7-11700K CPU @3.60 GHz and NVIDIA RTX3080ti. The operating system we use is Windows 10, and a deep learning framework is based on PyTorch (1.10.0). The optimizer used is adaptive moment estimation (Adam) [32], and the learning rate strategy is the "poly" strategy, with the following formula:

$$\text{lr} = \text{base_lr} \times \left(1 - \frac{\text{epoch}}{\text{num_epoch}}\right)^{\text{power}} \tag{17}$$

where lr is the updated learning rate, base_lr is the baseline learning rate, epoch is the quantity of iterations, num_epoch is the maximum number of iterations, and power controls the shape of the curve (usually it is greater than 1). In our model, power is set to 0.9, and num_epoch is set to 500. Because the training loss of most models in this experiment converges to 0 after 250 iterations, we fixed the epoch number to 300. We chose a batch size of 16 due to the graphics card's video memory constraint. The loss function used in this article is BCEWithLogitsLoss. As evaluation measures in this article, precision (P), recall (R), F_1 score, pixel accuracy (PA), mean PA (MPA), and mean intersection over union (MIoU) are used

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$F_1 = 2 \times \frac{\text{P} \times \text{R}}{\text{P} + \text{R}} \tag{20}$$

$$\text{PA} = \frac{\sum_{i=0}^{k} \rho_{i,j}}{\sum_{i=0}^{k} \sum_{j=0}^{k} \rho_{i,j}} \tag{21}$$

$$\text{MPA} = \frac{1}{k} \sum_{i=0}^{k} \frac{\rho_{i,j}}{\sum_{j=0}^{k} \rho_{i,j}} \tag{22}$$

$$\text{MIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{\rho_{i,j}}{\sum_{j=0}^{k} \rho_{i,j} + \sum_{j=0}^{k} \rho_{j,i} - \rho_{i,i}}. \tag{23}$$

The amount of accurately predicted building (water) pixels is represented by the true positive (TP) class. The incorrectly estimated building (water) pixel number is represented by the false positive (FP) class, and the true negative (TN) class represents the correctly classified nonbuildings (water) pixels, and false negatives (FNs) represent the number of misclassified building (water) pixels. k denotes the object segmentation class. The genuine class is represented by $\rho_{i,i}$, and $\rho_{i,j}$ denotes the number of pixels that relate to class i but are anticipated to belong to class j.

C. Ablation Experiments on Building and Water Datasets

Ablation experiments can demonstrate the impact of each module in our model. In this section, our model backbone uses the original ResNet50 structure as a baseline and upsamples all of its stages to obtain output. Following that, we incrementally integrate the proposed modules (DBMSA, MSF, and LCA) to the model to validate the viability of each module. The MIoU is utilized to evaluate network performance in this section, and the results are presented in Table II. It can be clearly seen

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TABLE II
ABLATION FOR DIFFERENT MODULES IN THE UNIT

Method	MIoU(%)
Backbone	86.41
Backbone+ASPP	86.57
Backbone+DBMSA	**86.83** (0.42 ↑)
Backbone+DBMSA+MSF	**87.14** (0.31 ↑)
Backbone+DBMSA+MSF+BAM	87.12
Backbone+DBMSA+MSF+DEF	87.34
Backbone+DBMSA+MSF+CBAM	87.80
Backbone+DBMSA+MSF+Non-local	87.82
Backbone+DBMSA+MSF+LCA	**88.23** (1.09 ↑)

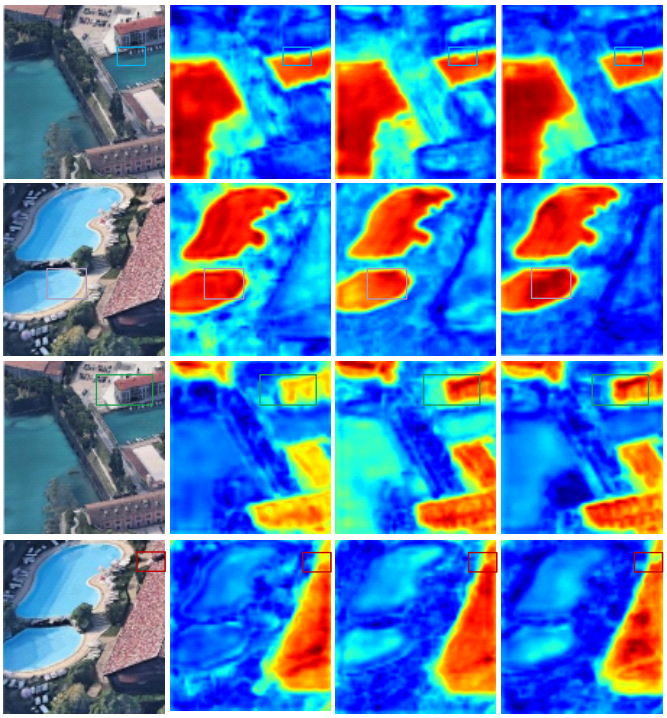

Fig. 10. Heatmaps of water and buildings for different modules. (a) Test image. (b) Method with DBMSA. (c) Method with DBMSA and MSF. (d) Method with DBMSA, MSF, and LCA.

from the table that the modules proposed in this article have a better improvement on this dataset.

We can intuitively see the effect of each module from the heat map. As illustrated in Fig. 10, the heat map shows the attention of the model to different classification areas. The red area receives the most attention from the model, the yellow–green area the second. The deeper the red, the greater the attention. The first and second rows of Fig. 10 represent the heat map of the module's attention to water, and the third and fourth rows of Fig. 10 represent the heat map of the module's attention to buildings. It can be seen from the heat map that the network is getting better and better for classification of buildings and water. After each module is added, the yellow–green area in the heat map will decrease, which means that the added modules can improve the network's attention to the segmentation target. In particular, our LCA module pays high attention to the details of the eaves and can accurately capture the details of objects. As shown by the green rectangles in the third row of Fig. 10, only the model with LCA has the highest attention to buildings, and the rectangles in Fig. 10(b) and (c) have yellow areas. This situation indicates that the two models are distracted on buildings. The red rectangle in the fourth row contains the shadows caused by the high and low eaves of the building. Shadows between houses are not counted in the building category, and we can see in Fig. 10(d) that only our model pays less attention to shadows. This also proves that our LCA is more focused on the classification region of interest, and its segmentation is better.

We know that CBAM and BAM are both channel spatial attention modules. The CBAM module is a lightweight convolutional attention module, which can effectively improve the network's attention to segmentation targets by concatenating CA and spatial attention. The BAM module is also a lightweight module, and it improves the network's attention to objects of interest by paralleling CA and spatial attention. The DFE module proposed by Hu is a location attention module, and it obtains location attention through the avg-pooling operation to obtain global information. Therefore, we compare the proposed LCA module with them. As shown in Table II, the model using the LCA module produces better results than the model using BAM, DFE, and CBAM. At the same time, we also performed ablation experiments with the original nonlocal neural networks. From the experimental results in Table II, it can be seen that the LCA unit proposed in this article was better than original nonlocal neural networks. This proved that the LCA unit can further improve the ability of the network to obtain long-distance dependencies of feature maps. This unit outperformed the existing hybrid attention modules and self-attention modules.

At the same time, in order to more intuitively reflect the advantages of the LCA module proposed in this article, we have visualized the heat map of the BAM, DFE, CBAM, and LCA modules, respectively. The first and second rows of Fig. 11 represent the heat map of the module's attention to water, and the third and fourth rows of Fig. 11 represent the heat map of the module's attention to buildings. Obviously, we can see that the attention given by the network in the column of Fig. 11(e) is concentrated on water and buildings, while the attention of other modules is more dispersed. The graphs of Fig. 11(b)–(d) in the fourth row have a large area of yellow and green, which indicates that the attention of these three networks is distracted by other objects, and the attention given to buildings is reduced. Meanwhile, we can clearly find from the fourth row of Fig. 11 that our LCA module pays less attention to the shaded regions of the eave edges that are not defined as labels. In contrast, the heat-map regions of other attention modules are redder in this part. This situation shows that these modules may have caused misjudgment in the shaded area. From the heat map in Fig. 11, we can clearly find that the LCA module proposed in this article is better than other attention modules, and the innovation of channel location attention is promising.

1) *Ablation Experiment of **DBMSA** Unit:* The improved atrous pyramid convolution pooling module is capable of extracting detailed multiscale features. We know

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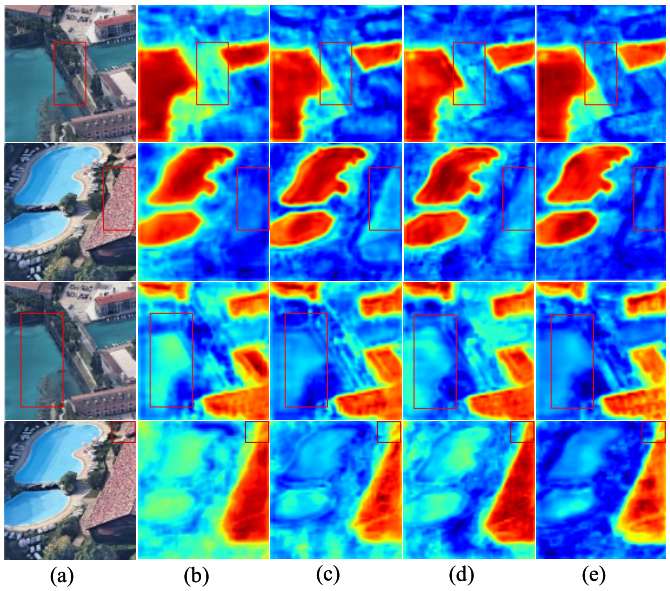

Fig. 11. Heat maps of waters and buildings for different modules. (a) Test image. (b) Method with BAM. (c) Method with DFE. (d) Method with CBAM. (e) Method with LCA.

that max pooling can extract informative features of images well, so we replace average pooling in ASPP. As can be seen from Table II, our proposed DBMSA produces better results compared with the ASPP module. The information fusion operation similar to the residual structure can complement the original feature map information, and the dual-branch output structure enables the network to obtain more exchange opportunities between deep information and low-level information. As shown in Table II, the DBMSA unit improves the MIoU value of the model from 86.41% to 86.83%, demonstrating that the unit gives correct classification predictions and extracts multiscale contextual features effectively.

2) *Ablation Experiment of **MSF** Unit:* This unit is used to merge high-level and low-level features, allowing high-level features with rich semantic information and low-level features with a lot of spatial features to guide each other, promoting the network's capacity to recognize and segment items. The refined convolution at the output can be optimized to generate predicted feature maps with insufficient detail after passing through the decoder. The data in Table II demonstrate that MSF is an efficient unit, which improves the MIoU of the unit by 0.31%.

3) *Ablation Experiment of **LCA** Unit:* The location CA unit can aggregate the feature map's location-spatial information and detail information. It enhances the model's capture of the detailed edge features through location attention and CA. The LSA unit can extract the spatial information of the feature map along the H- and W-directions, respectively. This method can enable the network to obtain large-scale features and prevent the overall information from being split while obtaining detailed information. We can see from Table II that this module can improve the model by 1.09%.

TABLE III
COMPARISON OF PARAMETER INDICATORS OF DIFFERENT CHANNEL NUMBERS OF THE MODEL

Channels A	Channels B	PA(%)	MPA(%)	MIoU(%)
256	512	**93.84**	**93.69**	**88.23**
256	1024	93.18	93.00	86.57
256	2048	93.25	93.01	86.62
512	1024	93.26	93.07	86.49
512	2048	93.23	93.12	86.63
1024	2048	93.00	93.08	85.93

TABLE IV
PERFORMANCE COMPARISON OF LCA WITH DIFFERENT REDUCTION RATES

Reduction Rate	PA(%)	MIoU(%)	Parameter(M)	Flops(G)
4	93.20	86.44	54.06	37.55
6	93.00	86.00	52.49	37.49
8	**93.84**	**88.23**	51.70	37.46
16	93.14	86.19	50.53	37.42
32	92.66	85.62	**49.94**	**37.40**

During the encoding process of our model, skip connections are not added to feature extraction layers in the backbone structure. The fundamental reason is that with the deepening of the feature extraction layer, the number of channels increases, which will lead to excessive model parameters and calculations, which will affect the running speed, and the accuracy will not improve much, so we only choose two places for skip connections. Because the feature information extracted from the first stem layer is too small, we do not consider selecting this layer to add skip connections. As shown in Table III, when the number of channels is 256 and 512, the PA, MPA, and MIoU of the model are the highest.

We found that skip connections greatly improved the prediction accuracy, but they also increase parameters and computation. According to studies, there would be redundancy in the amount of channels in the feature picture, so we can compress the channels. The channel reduction ratio r must be greater than or equal to 4 in order to effectively remove redundant channel features and reduce the amount of parameters. As illustrated in Table IV, we discovered that when $r = 8$, the model performs the best, and the number of parameters and computational complexity is appropriate.

D. Comparative Experiments on Different datasets

1) Comparison Test on Building and Water Datasets: We make a dataset from Google Earth to completely demonstrate that our algorithm can accurately classify buildings and water. This section chooses PA, MPA, and MIoU as the criteria. We selected sparse token transformers (STT) [33], global-feature capture and cross-layer information-interaction network (GCCINet) [34], RAPNet [35], deep dual-resolution network (DDRNet) [36], semantics guided bottleneck network (SGBNet), multiscale stripe pooling feature aggregation network (MSPFANet) [37], OCRNet [38], DeepLabV3+, MLNet, multilevel feature aggregation network (MFANet), and other excellent models for comparison. SegNet [39] and FCN adopt

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TABLE V
EVALUATION RESULTS OF DIFFERENT MODELS ON THE BUILDING AND WATER DATASETS

Method	PA(%)	MPA(%)	MIoU(%)	Flops($GMac$)	Params(M)	Time(s)
BiSeNetV2	89.29	90.14	79.58	2.45	3.62	169
SegNet	90.77	90.66	81.93	30.70	29.44	153
DDRNet	90.58	89.93	82.00	12.40	69.37	235
SGBNet	90.45	90.41	82.32	**0.20**	**0.05**	185
MSPFANet(ResNet18)	91.18	92.21	83.83	28.46	33.7	201
FCN8s	91.91	91.78	84.47	62.69	134.28	167
OCRNet(ResNet101)	92.60	91.48	84.55	30.93	70.35	676
ENet	92.04	92.70	85.03	0.34	0.34	223
DFN(ResNet101)	92.21	93.27	85.06	7.81	42.53	285
UNet	92.82	92.93	85.67	30.63	17.27	**117**
DANet(ResNet50)	92.82	92.86	86.09	37.76	62.01	178
PSPNet(ResNet101)	92.66	92.33	86.28	50.13	68.06	257
DeepLabV3+(ResNet101)	92.71	92.35	86.44	49.70	91.77	273
ACFNet(ResNet50)	93.03	93.50	86.47	76.69	89.97	282
MLNet(VGG16)	93.33	93.20	86.81	35.80	22.84	208
MFANet(ResNet50)	93.42	93.89	86.97	21.56	80.85	306
CCNet(ResNet50)	93.25	93.86	87.06	42.63	52.28	271
RAPNet(ResNet101)	92.84	93.16	86.31	60.24	70.35	278
GCCINet	93.28	93.30	87.00	33.24	25.78	224
STT(ResNet50)	93.23	93.30	87.01	52.34	18.75	245
MSLANet	**93.82**	**94.29**	**88.09**	28.71	51.70	257

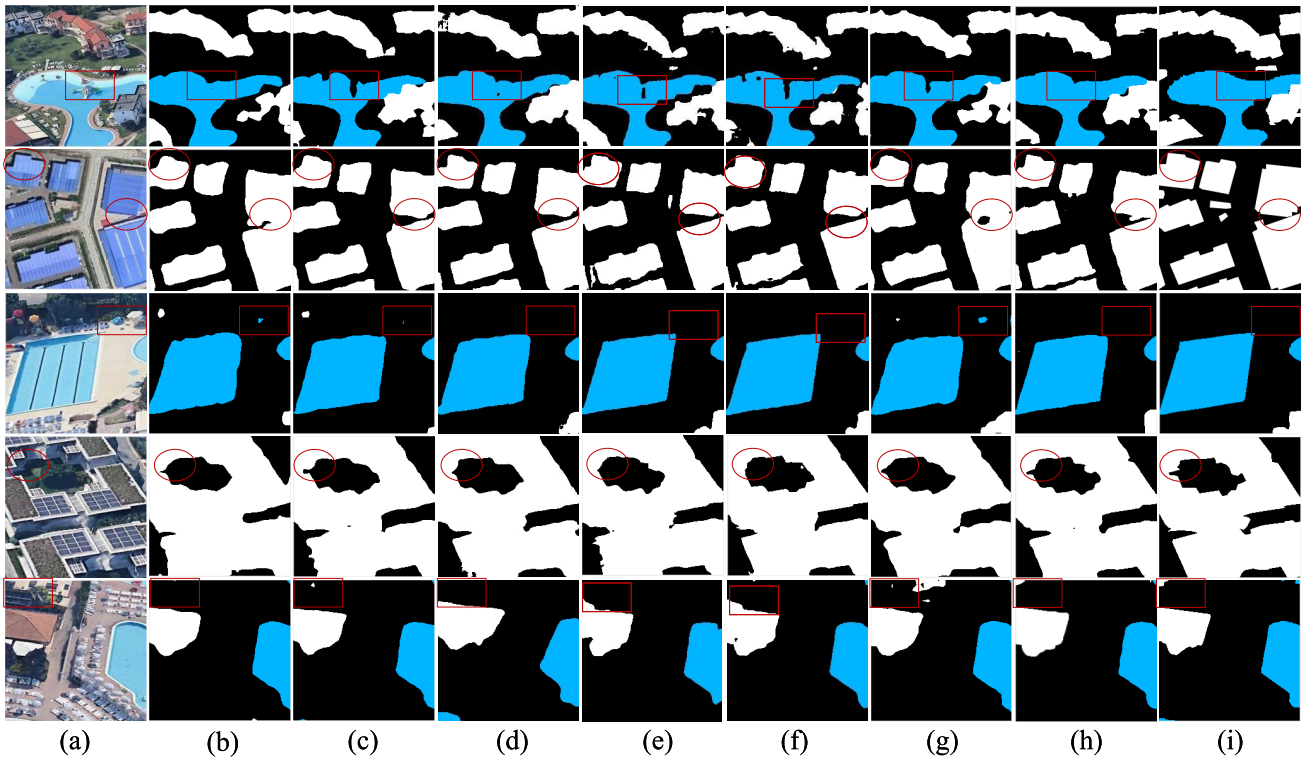

Fig. 12. Comparison results on the building and water datasets. (a) Original images. (b) Predicted maps of ACFNet. (c) Predicted maps of MLNet. (d) Predicted maps of MFANet. (e) Predicted maps of GCCINet. (f) Predicted maps of STT. (g) Predicted maps of CCNet. (h) Predicted maps of MSLANet. (i) Corresponding labels.

the structure corresponding to the encoder and decoder and the fully convolutional structure to achieve pixel-level semantic classification, respectively. ENet [40] is a real-time semantic segmentation network based on the idea of ResNet. Dual attention network (DANet) [41] uses both spatial attention and CA to enhance the image segmentation. Attentional class feature network (ACFNet) [42] adopts the concept of class center, which highlights the category context information and achieves accurate classification. DeepLabV3+ uses the ASPP module composed of various atrous convolutions to extract image multiscale information, and PSPNet uses the PPM module to achieve multiscale extraction. UNet first proposes a "U"-like encoding and decoding structure, which effectively realizes accurate segmentation. Discriminative feature network (DFN) [43] adopts global pooling operation, improves the CBA CA module, and enhances the information capture ability. Criss-cross attention network (CCNet) [44] proposes criss-cross attention module, which reduces the computational complexity of the semantic segmentation network. DDRNet is a segmentation network that uses dual-branch

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channels to extract contextual semantic information and retain detail information. Object-contextual representations network (OCRNet) [38] focuses on using target region representation to enhance the representation of segmentation target pixels. BiSeNetV2 [45] is a lightweight semantic segmentation network that uses a bilateral-guided aggregation (BGA) layer to enhance interconnectivity and fuse two types of feature representations. The abovementioned networks are widely used in semantic segmentation. STT, GCCINet, refined attention pyramid network (RAPNet), SGBNet, MSPFANet, MFANet, and MLNet are networks dedicated to remote sensing image segmentation. Among them, MSPFANet is a network widely used for remote sensing image segmentation. It uses strip convolution to improve the PPM module to extract multiscale information of the network. SGBNet, MLNet, and MFANet are models dedicated to building and water segmentation. SGBNet is a lightweight model. This model proposes a bilateral feature aggregation method that effectively combines the results of bidirectional feature extraction. MLNet and MFANet, respectively, propose a weighted index attention mechanism and a DFE module, both of which are guided by the attention mechanism to achieve accurate segmentation of buildings and waters. The STT is a Transformer variant network with a dual-path transformation structure. GCCINet is a cross-layer information interaction network that uses dilated convolutions to extract multiscale architectural feature information. RAPNet is an attention-based pyramid network that uses PPMs, dilated convolution modules, and deformable convolution modules to improve network feature extraction capabilities. The computational complexity (flops) of the network, the number of parameters (params), and the training time (time) of the dataset can reveal the time efficiency of the model. The time efficiency of the model greatly shows the possibility of realistic deployment of the network. The parameter quantity, computational complexity, and dataset training time of the 21 networks are shown in Table V. From the table, we can see that the UNet has the shortest training time, because its structure is less complex. However, its various segmentation indicators are poor, and it cannot well handle building and water segmentation. In addition, the parameter amount and computational complexity of SGBNet are the lowest among the 21 networks, but its segmentation is poor. Although the network in this article is higher than the lightweight model in terms of computational complexity and parameter volume, in these two indicators, we are still better than non-lightweight models. In addition, the training time of our network in this article is also in the top-middle position of the 18 networks. As shown in Table V, MSLANet achieves the highest scores on three important metrics (PA/MPA/MIoU: 93.82%/94.29%/88.09%). Therefore, our proposed network performs the best with little compromise in computational complexity, data volume, and training time.

In Fig. 12, we selected seven better-performing networks among the 20 control networks to demonstrate their segmentation performance. The MSLANet network can effectively avoid the misclassification caused by the similar pixel values of some buildings and water, and the interfering situation that there is a hollow background inside the building. As illustrated by the red rectangle in the third row of Fig. 12, the rest of the control networks misclassify umbrellas as buildings and blue chairs as water, but our network did not misclassify. For the hollow inside the building, the segmentation is more difficult. In the ellipse box in the fourth row of Fig. 12, our network also performs better than the control network in the detailed edge segmentation of the hollow background. At the same time, the segmentation of the water area is also more difficult. As shown in the rectangular box in the first row of Fig. 12, there are slides, floating toys, and other objects in the swimming pool class, which will cause the network to misclassify it as the background. However, our model in this article can segment the water area very well without being disturbed. Our model can well combine the detail information and the location space information, which benefits from our location CA module. Therefore, in Fig. 12, our network captures the edge details of buildings and water better. In addition, our network will not lose too much global information of buildings and water, avoiding the phenomenon of fragmenting the overall information like other networks.

TABLE VI
COMPARISON OF EVALUATION METRICS OF DIFFERENT MODELS ON THE BUILDING AND WATER DATASETS

	Building			Water		
Method	P(%)	R(%)	F_1 (%)	P(%)	R(%)	F_1 (%)
BiSeNetV2	85.58	68.58	76.24	95.12	95.56	95.34
SegNet	85.04	76.68	80.64	94.95	94.90	94.93
DDRNet	83.57	78.65	81.04	93.80	96.35	95.06
SGBNet	82.22	79.10	80.63	96.61	95.43	96.02
MSPFANet(ResNet18)	87.98	78.30	82.86	96.28	95.12	95.70
FCN8s	84.87	82.61	83.73	96.81	95.29	96.04
OCRNet(ResNet101)	87.10	82.94	84.97	92.99	95.87	94.41
ENet	87.94	77.89	83.17	97.59	97.03	97.31
DFN(ResNet101)	89.51	78.13	83.43	98.03	96.02	97.01
UNet	89.11	81.50	85.14	96.13	96.12	96.13
DANet(ResNet50)	87.59	83.41	85.45	96.98	96.28	96.63
PSPNet(ResNet101)	87.20	84.12	85.22	97.23	97.05	97.12
DeepLabV3+(ResNet101)	84.67	**86.32**	85.61	97.34	95.60	96.77
ACFNet(ResNet50)	**90.38**	80.71	85.28	96.75	96.61	97.20
MLNet(VGG16)	87.82	85.83	86.81	97.06	96.26	96.66
MFANet(ResNet50)	90.12	83.46	86.67	96.62	95.68	96.63
CCNet(ResNet50)	89.19	83.25	86.12	**98.48**	96.28	97.37
RAPNet(ResNet101)	87.91	83.58	85.09	97.01	95.56	97.27
GCCINet	88.82	84.13	86.41	96.81	96.01	97.04
STT(ResNet50)	88.93	84.70	85.85	98.43	96.35	97.33
MSLANet(Our)	89.86	86.10	**87.43**	**98.48**	**96.63**	**97.55**

Table VI compares the results different networks. On both R and F_1, our network outperforms the existing networks in building detection and water detection. We can see that in the classification of buildings, the precision (P) score of ACFNet is the highest, mainly because the network adopts an attention network for classification, which can adaptively segment corresponding categories according to pixels, but we also can see from Fig. 12 that its segmentation accuracy is not very good. Although our approach's accuracy in the building category is not the highest, it is only a little bit lower than the best performing method. On the segmentation dataset of buildings and water, our network has nearly the highest scores in all three indicators, reaching 89.86%, 86.10%, 87.43%, 98.48%, 96.63%, and 97.55%, respectively, and our segmentation accuracy is also the best.

In order to further verify that the model in this article still also has a strong segmentation ability in some special cases,

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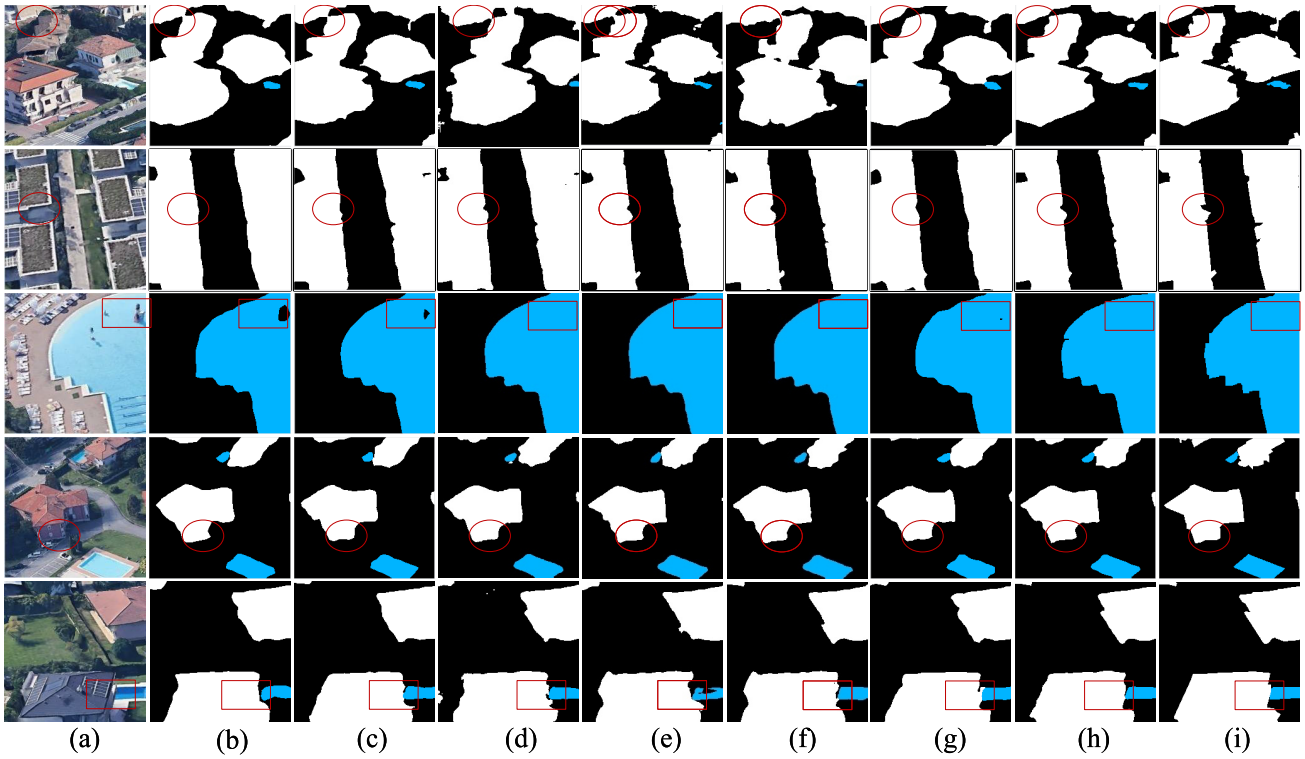

Fig. 13. Comparison results on the building and water datasets. (a) Original images. (b) Predicted maps of ACFNet. (c) Predicted maps of MLNet. (d) Predicted maps of MFANet. (e) Predicted maps of GCCINet. (f) Predicted maps of STT. (g) Predicted maps of CCNet. (h) Predicted maps of MSLANet. (i) Corresponding labels.

we selected pictures containing complex scenes in Fig. 13 and showed the training effect under different models. For example, a building has hollows, there are floating objects on the water, in all blue-roofed house and the water are similar in pixels, and view angles of the dataset change greatly. These situations are likely to make it difficult for the network to detect the target object. Therefore, we selected pictures containing the abovementioned complex scenes in Fig. 13 and showed the test results under different models. As demonstrated in the second and third rows of Fig. 13, there are green plants in the buildings and objects, such as lifebuoys and observation platforms on the water. In such case, other comparison networks have misclassified situations, whereas our model does not misclassify the observatory on the water as the background. Also, in the rectangular box in the fifth row of Fig. 13, because the building is blue, it has similar pixel values to the water area; CCNet and other networks mistakenly classify part of the water area as a building. Our network not only generates such error classification, but also performs well in processing the detailed extraction of house edges and swimming pool edges. This situation is due to the insufficient extraction of multiscale information and the lack of sufficient spatial information. Therefore, this article proposes a DBMSA unit to improve the ability of multiscale features extraction. Similar to the label image in Fig. 13, for example, in the oval box in Fig. 13 of the first row and the fourth row, our method provides the best results in the recognition of the intricate swimming pool edges, hollow background edges, and building eaves. All these improvements benefit from the improved spatial CA module. By effective global attention along the H- and W-directions, a large amount of global spatial information could be obtained. This operation effectively avoids misclassifying observation towers on the water as background, and the sun umbrellas and sun loungers as buildings. At the same time, the CA module effectively extracts the edge information of buildings and water. Compared with ACFNet, MLNet, MFANet, GCCINet, and STT in the first row of Fig. 13, where the segmentation edges are blurred and rough, our model effectively extracts the details of buildings and edges and has better segmentation. Of course, the segmentation of MLNet, GCCINet, STT, and MFANet is also relatively good, mainly because they are also networks specialized in building and water segmentation. But, compared with our MSLANet, they still have some shortcomings. It can be seen from the first row of Fig. 13 that the shadow of the house is classified as the background, or as in the second row of Fig. 13, the detection of the house is incomplete, resulting in misclassification. At the same time, as shown in the third row of Fig. 13, MFANet also has misclassification. The experimental results show that our method can achieve accurate segmentation under above interference, and it greatly reduces misclassification.

We also perform generalization experiments on the five-class L8SPARCS dataset and the two-class water dataset to further validate the model's effectiveness.

2) Generalization Experiment of the L8SPARCS Dataset: As can be seen in Table VII, MSLANet has the highest scores in PA, MPA, and MIoU, reaching 92.67%, 88.93%, and 81.61%, respectively. Compared with the control network, MSLANet is also the best on the L8SPARCS dataset, and its MIoU is improved by at least 2.45%. This experiment shows that MSLANet has a strong generalization ability.

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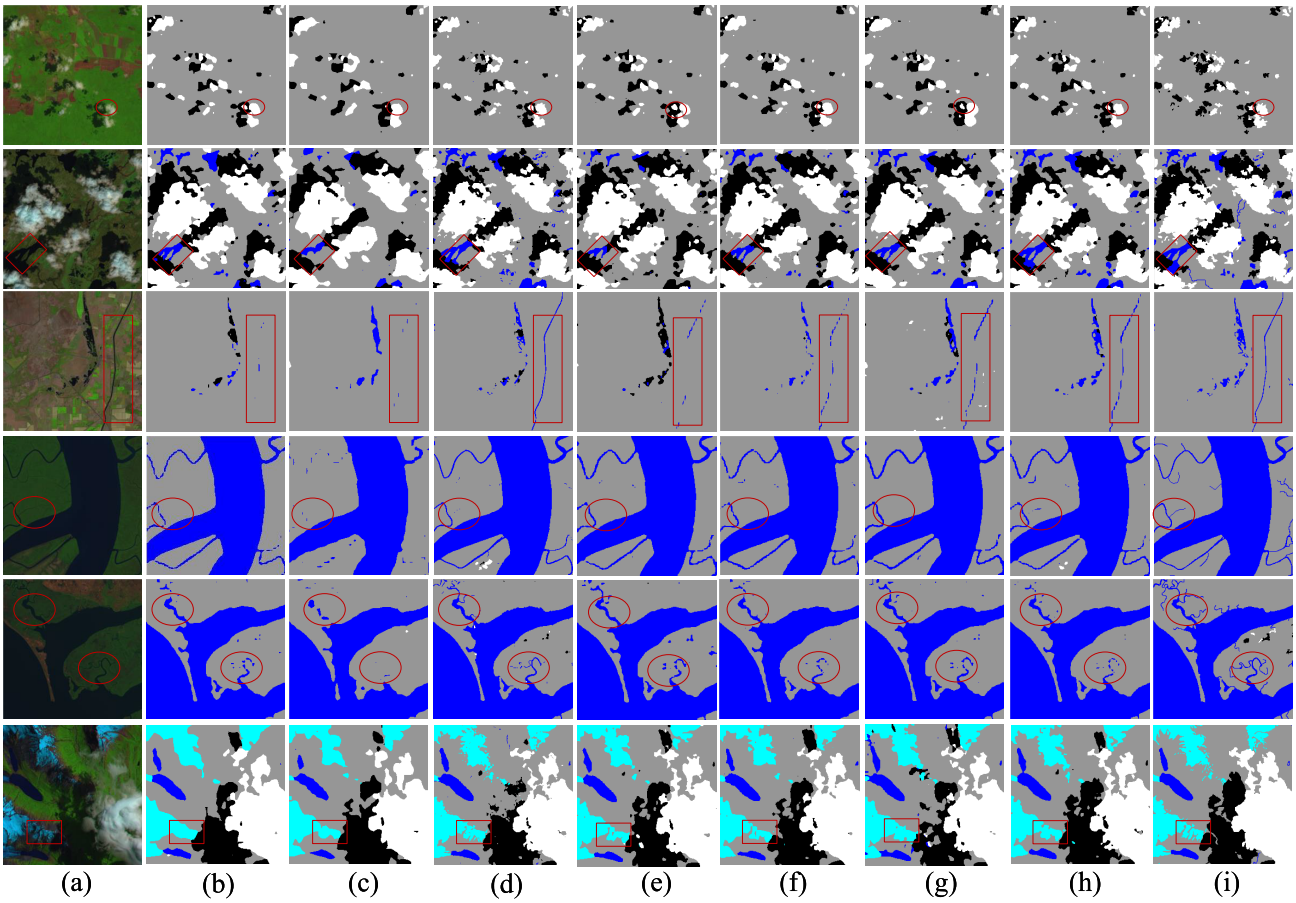

Fig. 14. Comparison results on the L8SPARCS dataset. (a) Original images. (b) Predicted maps of DeepLabV3+. (c) Predicted maps of ACFNet. (d) Predicted maps of UNet. (e) Predicted maps of GCCINet. (f) Predicted maps of OCRNet. (g) Predicted maps of STT. (h) Predicted maps of MSLANet. (i) Corresponding labels.

TABLE VII
EVALUATION RESULTS OF DIFFERENT MODELS ON THE L8SPARCS DATASET

Method	PA(%)	MPA(%)	MIoU(%)
BiSeNetV2	86.53	81.05	70.12
SegNet	89.64	83.43	73.05
DDRNet	90.27	83.34	74.72
SGBNet	88.83	81.72	69.07
MSPFANet(ResNet18)	89.14	85.04	72.71
FCN8s	90.66	86.51	75.48
OCRNet(ResNet101)	92.11	86.19	78.56
ENet	91.35	85.38	77.14
UNet	91.56	88.41	77.94
DANet(ResNet50)	90.62	83.00	74.90
PSPNet(ResNet101)	91.54	84.59	77.02
DeepLabV3+(ResNet101)	91.58	84.71	77.39
ACFNet(ResNet50)	91.62	84.60	77.53
MLNet(VGG16)	90.95	87.77	76.10
MFANet(ResNet50)	91.16	87.36	76.52
CCNet(ResNet50)	91.37	84.76	76.80
RAPNet(ResNet101)	90.87	87.51	77.07
GCCINet	91.51	87.47	78.33
STT(ResNet50)	92.17	87.81	79.16
MSLANet(Our)	**92.67**	**88.93**	**81.61**

The segmentation performance of MSLANet on the five-category dataset is also superior than that of other networks, as demonstrated in Fig. 14. If a small target with complex edge features can be separated from a complex background, the performance of the model is very good. MSLANet can capture the details of clouds and cloud shadows, as well as water tributaries against complicated backgrounds, as shown in the rectangular box in the third row of Fig. 14. Even though rivers and cloud shadows are similar in pixel values, the model can easily classify rivers and cloud shadows. It can be seen that the ability of other networks to detect rivers is relatively poor. The information in the segmented image is seriously lost, and there is no outline of the river. Only UNet and MSLANet can roughly detect the outline of the river, but it can be clearly seen that UNet has misclassified the rivers. MSLANet relies on the LCA unit to extract detailed information and spatial information, effectively eliminating the interference of cloud shadows on river in multiclassification. Low-level spatial features and high-level semantic features are bidirectionally aggregated with the assistance of the MSF unit, and this procedure is very effective in edge detection of clouds, cloud shadows, and water. Therefore, MSLANet is very accurate in capturing the detailed information of the detected objects and will not lose global information and cause fragmentation. This is benefited from our DBMSA unit, which can capture multiscale information to complete object details through multiscale module information.

3) Generalization Experiment of the Water Dataset: The water dataset is a binary dataset for water and background. The water of this dataset has the following: 1) different

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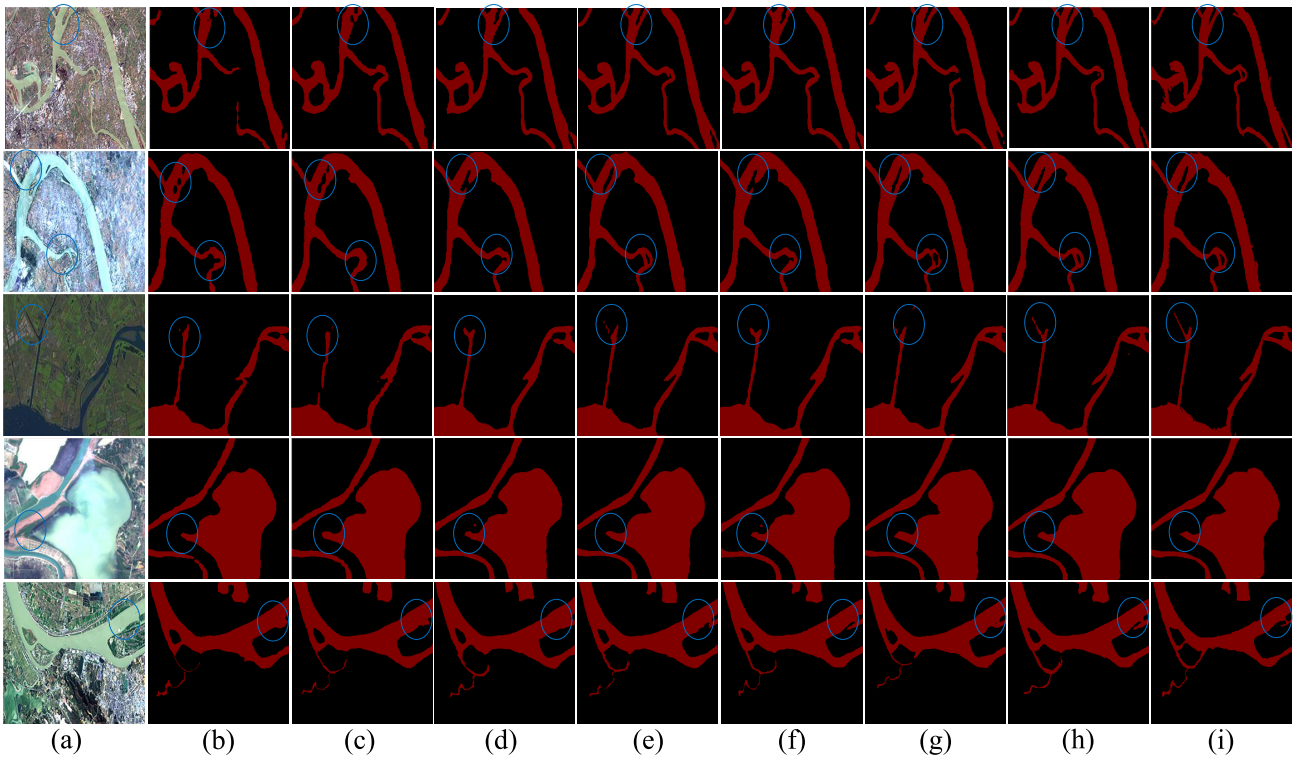

Fig. 15. Comparison results on the water dataset. (a) Original images. (b) Predicted maps of ACFNet. (c) Predicted maps of BiSeNetV2. (d) Predicted maps of GCCINet. (e) Predicted maps of OCRNet. (f) Predicted maps of MFANet. (g) Predicted maps of STT. (h) Predicted maps of MSLANet. (i) Corresponding labels.

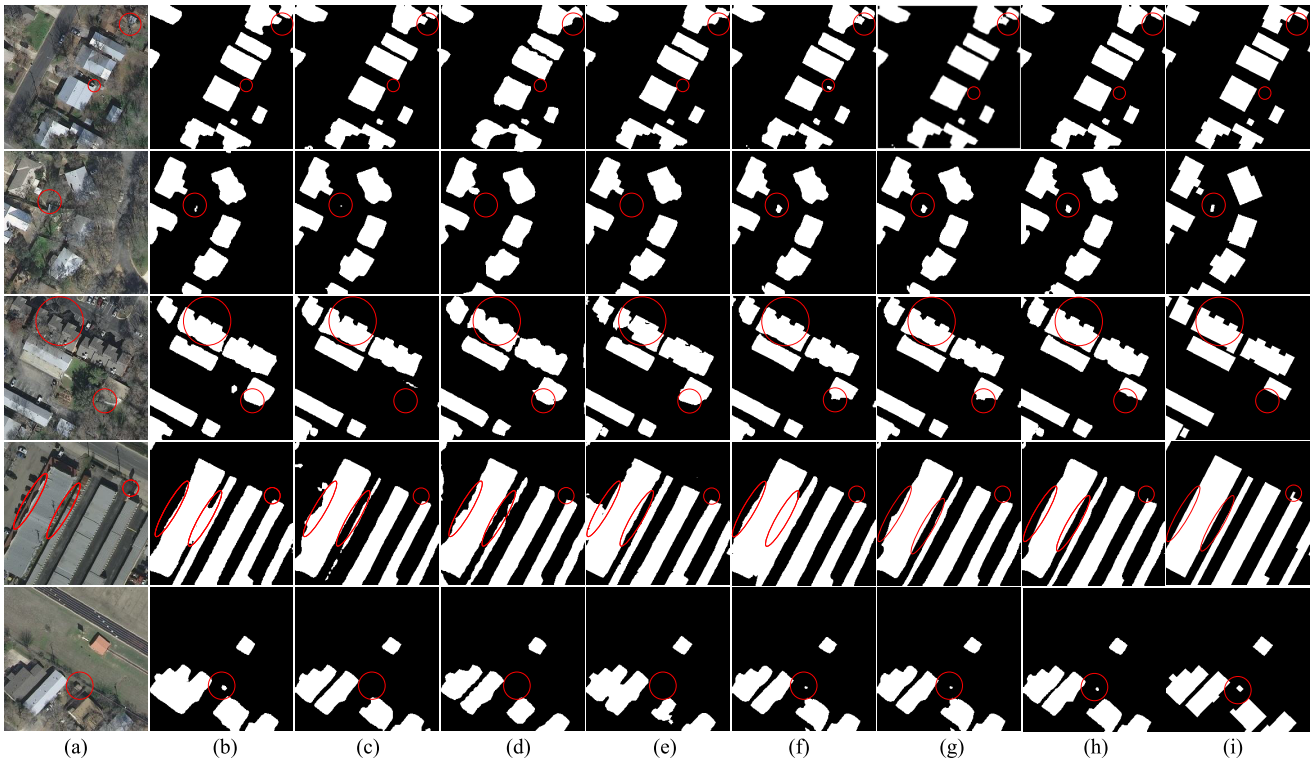

Fig. 16. Comparison results on the Inria dataset. (a) Original images. (b) Predicted maps of ENet. (c) Predicted maps of RAPNet. (d) Predicted maps of ACFNet. (e) Predicted maps of UNet. (f) Predicted maps of GCCINet. (g) Predicted maps of STT. (h) Predicted maps of MSLANet. (i) Corresponding labels.

water scenarios, such as rivers in cities, rivers and lakes in forests, and snow backgrounds; 2) selected water information covering early, middle, and late three time periods; and 3) also contains a lot of details, for example, the tributaries of the river have a lot of details, and there are inland islands in the river. Therefore, whether the detailed tributaries of the

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TABLE VIII
EVALUATION RESULTS ON THE WATER DATASET

Method	P(%)	R(%)	F_1 (%)	IoU(%)
BiSeNetV2	97.81	95.16	96.46	93.17
SegNet	97.32	94.86	96.07	92.44
DDRNet	97.65	95.14	96.38	93.01
SGBNet	96.26	94.37	95.31	91.03
MSPFANet(ResNet18)	97.02	94.90	96.38	93.01
FCN8s	97.29	93.82	95.52	91.42
OCRNet(ResNet101)	97.75	95.52	96.62	93.46
ENet	97.36	94.46	95.89	92.10
DFN(ResNet101)	97.09	94.75	95.91	92.14
UNet	97.73	94.68	96.26	92.80
DANet(ResNet50)	97.13	94.63	95.87	92.06
PSPNet(ResNet101)	97.05	95.50	96.26	92.80
DeepLabV3+(ResNet101)	97.04	94.94	95.98	92.28
ACFNet(ResNet50)	97.78	95.12	96.43	93.11
MLNet(VGG16)	97.52	95.33	96.54	93.02
MFANet(ResNet50)	97.77	95.72	96.73	93.67
CCNet(ResNet50)	97.34	94.71	96.00	92.33
STT(ResNet50)	97.32	96.24	96.78	93.76
RAPNet(ResNet101)	96,29	95.92	96.10	92.50
GCCINet	97.01	96.02	96.52	93.27
MSLANet(Our)	**97.86**	**96.28**	**97.06**	**94.30**

TABLE IX
EVALUATION RESULTS ON THE INRIA DATASET

Method	P(%)	R(%)	F_1 (%)	IoU(%)
BiSeNetV2	87.20	83.33	85.22	74.24
SegNet	86.25	84.59	85.41	74.54
DDRNet	87.37	84.50	85.91	75.30
SGBNet	87.34	82.62	84.91	73.78
MSPFANet(ResNet18)	87.96	83.30	85.57	74.77
FCN8s	86.47	84.11	85.28	74.33
OCRNet(ResNet101)	86.80	87.21	87.00	76.54
ENet	87.34	86.31	86.82	76.71
DFN(ResNet101)	86.75	82.42	96.25	73.20
UNet	87.65	86.29	86.96	76.93
DANet(ResNet50)	87.64	84.86	86.23	75.79
PSPNet(ResNet101)	86.89	86.79	86.84	76.75
DeepLabV3+(ResNet101)	87.81	85.23	86.50	76.21
ACFNet(ResNet50)	88.49	85.34	86.89	76.82
MLNet(VGG16)	87.01	83.35	85.11	73.45
MFANet(ResNet50)	87.51	85.37	86.43	76.10
CCNet(ResNet50)	88.30	84.73	86.48	76.18
RAPNet(ResNet101)	87.76	85.30	86.81	76.78
GCCINet	88.97	87.30	87.89	78.79
STT(ResNet50)	89.14	87.42	87.91	79.35
MSLANet(Our)	**89.21**	**87.56**	**87.93**	**79.41**

water area can be accurately detected reflects the ability of the model.

Because the water dataset is a binary classification dataset, we use P, R, F_1, and IoU, as the evaluation indicators. Table VIII shows that compared with other control networks, MSLANet performs better in detecting the water dataset. In the four indicators of P, R, F_1, and IoU, MSLANet reaches 97.86%, 96.28%, 97.06%, and 94.30%, respectively, which is at least 0.54% higher than other networks in this indicator.

In addition to the improvement in accuracy, it is obvious from the ellipse box in the third line of Fig. 15 that the tributaries of the river are the difficulties in the detection of this dataset. Only MSLANet recognizes the tributaries of the river in the top-left corner, whereas the other networks could not. In addition, MSLANet can detect the junction of rivers and buildings clearly, which can accurately detect and segment the river contour we need. With the LCA unit, MSLANet can clearly capture the tributaries of rivers and the boundary of rivers and buildings. At the same time, because of the global attention module in LCA, the feature map will not lose the global spatial information even there is too much attention to detail information. After the feature map has passed through the backbone network, the DBMSA unit can mine the image's deep semantic information further, helping the model to obtain more multiscale information of the river, which is beneficial to the final detection. The multiscale aggregation module (MSF) can effectively aggregate the overall features of low-level rivers and the detailed features of high-level rivers, allowing the model to summarize the overall characteristics of the image. This module can not only extract the detailed information of tributaries and boundaries, but also avoid of misclassification caused by splitting. Finally, the refined convolution module of MSF can further optimize our predicted river contour information, complete the detailed information, and better improve the detection results.

4) Generalization Experiment of the Inria Dataset: Generalization Experiment of the Inria Dataset: The Inria dataset is a binary classification dataset of buildings and backgrounds. The images in this dataset come from different cities and regions, including different types of areas, such as urban, suburban, and rural. These areas contain a wide variety of buildings, which thoroughly test the segmentation ability of the model. Table IX shows that the scores of MSLANet on the Inria dataset on P, R, F_1, and IoU are 89.21%, 87.56%, 87.93%, and 79.41%, respectively. The network in this article also performs better than other state of the arts (SOTA) methods on the Inria dataset.

By testing on the public dataset Inria dataset, we can find that the segmentation performance of the MSLANet network is also excellent, as shown in Fig. 16. At the same time, this dataset contains a large number of rural houses. This type of house usually has a prominent eaves, and the eaves are classified as buildings. Therefore, it is critical to accurately capture the overall information of the building and improve the eaves detail information of certain houses. As shown in Fig. 16, we can see that only our network accurately captures the overall information of the house and does not classify the debris stacked next to the house into a category of buildings. It can be seen from the round box of the second line of Fig. 16 that error classification occurred in the network, such as UNet and ACFNet, and they ignore the segmentation information of a small house. We can see from the round frame of the third line in Fig. 16 that MSLANet's capture of the eaves of the house is very detailed. On the other hand, when the UNet and GCCINet dealt with edge details, their segmentation was not as good as our network. As shown in the fourth row of the oval box, the slender buildings are similar to the warehouse, and the network classifies the empty space in the middle as a building. MSLANet not only classified the space correctly, but also can capture the details of the edge of the building. The main reason for the above problems is that certain networks cannot

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effectively obtain the multiscale information of the feature map and context information. The DBMSA unit proposed in this article can effectively extract multiscale information. As shown in the round frame of the fifth line of the Fig. 16, UNet, ACFNet, and other networks had mistakes. Our method can enhance the extraction of multiscale information, enabling the network to better adapt to different resolutions and reduce the possibility of wrong classification. At the same time, our network benefits from the position attention of the LCA unit. It can increase the network's target attention to the eaves of the building and enhance the segmentation on the edge. The end refinement of the MSF unit can also optimize restoring edge details during the sampling process. To conclude, the segmentation ability of the MSLANet proposed was the best, and its generalization ability is strong.

IV. Conclusion

Based on high-resolution remote sensing photographs, this study presents a multiscale location-attention network for end-to-end building and water detection. The method takes the improved ResNet50 as the backbone, mines semantic information at different stages, and uses the DBMSA module to further extract multiscale semantic features. Next, the network extracts features from the encoding layer with channel numbers 256 and 512 through skip connection and then inputs the feature information into the location CA module. Through this method, the network can extract sufficient global spatial information and detailed local information, so that the two kinds of information complement each other and improve the information features. Finally, the MSF module guides the fusion of high-level semantic information and low-level spatial information of the attention module and the DBMSA module. The thinning convolution at the tail redetects the target object boundary and continues to repair the original rough boundary. Through the above operations, the boundary of the output feature map will be better improved, and the segmentation effect of the target object will be better. However, this method still could be further improved in the detection of buildings and water, and its parameter quantity and computational complexity can also be optimized. On the premise of ensuring the segmentation performance, it is vital to decrease the amount of parameters and computational complexity, as well as training time.

Acknowledgment

The computer code is available at https://github.com/Kimmadx/Kimma.

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